Influence of Animacy and Collectivity on the Production of the Optional Mandarin Plural Marker *men* (们)

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Introduction

Previous Findings

- *Men* is not obligatory [1]; mostly [+human] [2]
- Marks plurality/collectivity [5]; sensitive to discourse, politeness, and identity [3][4]

Research Gap

- Prior work mostly introspection + corpus
- PI, PG, and noun types not adequately distinguished

Research Focus

- Primary: native speakers' production of *men*
- Noun type (H/A/O) & quantity (S/PG/PI)
- Secondary: interaction with other plural strategies (*ji3ge2* "several", etc.)

Materials & methods

Participants

- 84 native Mandarin Chinese speakers (age:18-32)

Procedure

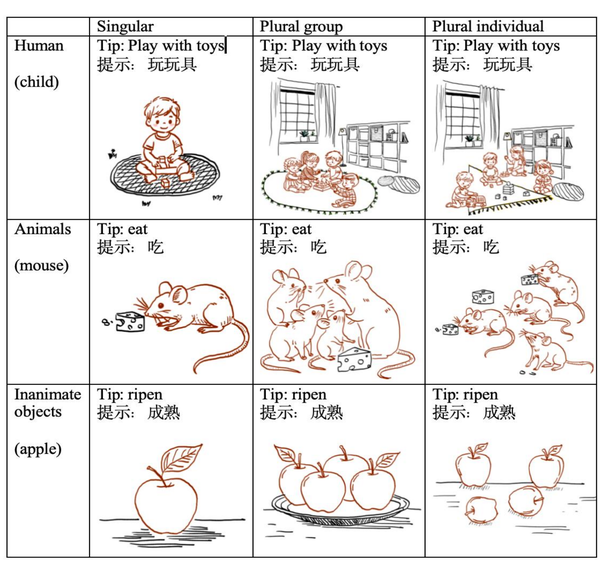
- 27 full-sentence picture descriptions per participant
- in-person & online meeting

Transcription & coding

- Audio recorded; orthographically transcribed
- Annotations for target noun
- Coding *men* and other number-marking strategies

Table 1. Experimental image examples.

	Singular	Plural group	Plural individual
Human (child)	Tip: Play with toys 提示：玩玩具	Tip: Play with toys 提示：玩玩具	Tip: Play with toys 提示：玩玩具
Animals (mouse)	Tip: eat 提示：吃	Tip: eat 提示：吃	Tip: eat 提示：吃
Inanimate objects (apple)	Tip: ripen 提示：成熟	Tip: ripen 提示：成熟	Tip: ripen 提示：成熟



Results

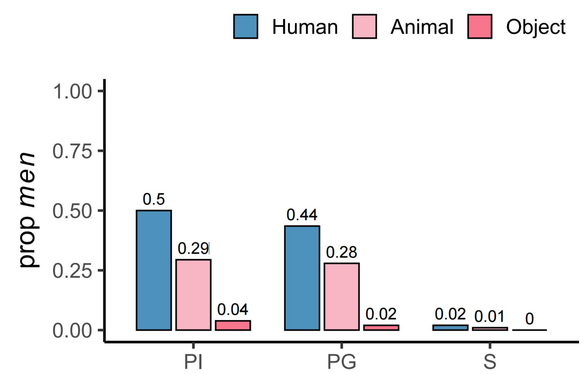


Fig 1. *Men* marking across noun types and number conditions.

Table 2. Estimates for the mixed-effects model predicting the probability of *men* production (plural trials only).

Fixed effects	Estimate	Std. Error	z value	Pr(>|z|)
Intercept	-3.6441	0.5734	-6.356	***p < .001
noun type: animal vs human	-2.4186	0.4357	-5.551	**p < .01
noun type: object vs animal	-5.6378	0.7318	-7.704	*p < .05
quantity	0.5531	0.3171	1.744	0.0811 .
animal-human x quantity	-0.3708	0.5380	-0.689	0.4907
object-animal x quantity	0.6453	0.8627	0.748	0.4544

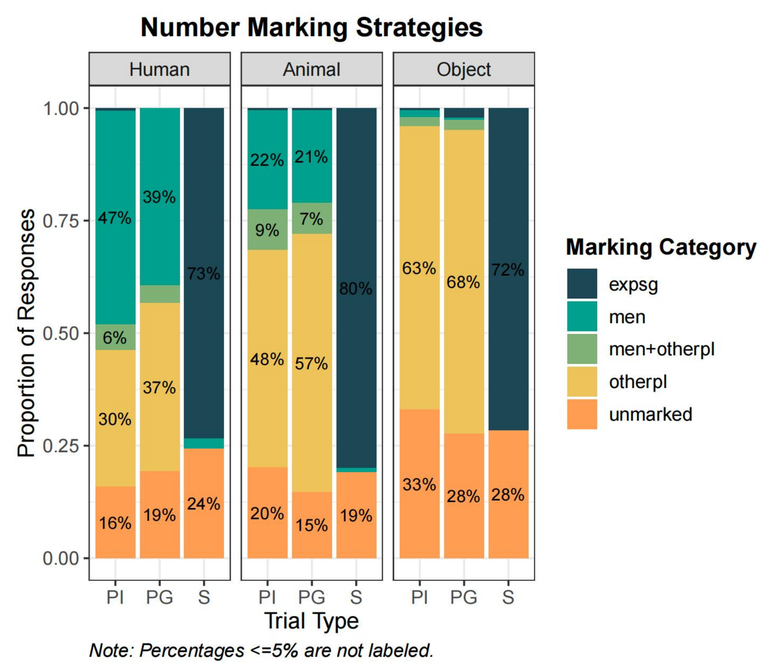


Fig 2. Proportional distribution of different number marking strategies.

- ***Men***: mainly **human**, sometimes **animal**, rare with **objects** (Fig 1)
- ***Men***: almost only in **plurals** (PI, PG) (Fig 1)
- **PI** vs **PG** variation < **noun type** effect (Table 2)
- **PI** vs **PG** differences ≈ no sig. effect on ***men*** (Table 2)
- Baseline ***men*** usage low (Table 2)
- **Other plurals** dominate **non-human nouns** (Fig 2)
- **Expsg** dominates in **singular** (Fig 2)
- **Unmarked** is low for **human** & **animal** nouns (Fig 2)
- **Plurals**: *men* (human) / otherpl (animal/object); **singulars**: expsg (Fig 2)

Discussion

- ***Men*** is the primary plural marker studied, forming a flexible and optional system
- ***Men*** shows strong sensitivity to **plurality** and at most weak sensitivity to **collectivity**
- ***Men*** coexists with **other plural** strategies, showing variability in plural marking
- System favors humans but permits ***men*** for **animals**
- Singular marking strategies (expsg, unmarked) interact with plural marking to create a **balanced system**

References

[1] Cook, A. (2009). Why should *men* break all the rules? A new approach to analyzing the plural marker *men* in Mandarin Chinese. Linguistics & the Human Sciences, 5(2).
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[5] Hewitt, S. T. (2012). Modalising plurals. Journal of Philosophical Logic, pp. 41, 853–875

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